

Advanced Machine Learning 2026/2027

Project: **Modeling Recovery after Lower-Limb Musculoskeletal Injury**

1. Introduction

Recovery following a lower-limb musculoskeletal injury varies substantially across patients. Differences in injury severity, age, physical condition, comorbidities, and adherence to rehabilitation can influence both the pace and extent of recovery. During rehabilitation, healthcare professionals monitor indicators such as mobility, pain, fatigue, and activity, and adapt rehabilitation interventions as the patient's condition evolves.

A patient may initially have limited mobility and substantial pain, gradually regain function throughout rehabilitation, or experience periods of slower recovery or temporary setbacks. Decisions about the type and intensity of rehabilitation must therefore be made repeatedly over time, while the patient's underlying recovery condition is only indirectly reflected by observable measurements. Modeling recovery under uncertainty and supporting sequential decision-making are important challenges in rehabilitation.

In this project, you will step into the role of a data scientist analyzing longitudinal data from patients undergoing rehabilitation following a lower-limb musculoskeletal injury. Patients are followed over several weeks, during which functional and patient-reported measurements are recorded and different rehabilitation interventions may be applied. Your task is to build models that can:

- **Identify latent patterns** in patient rehabilitation data.
- **Represent and reason about probabilistic relationships** among patient characteristics, recovery states, measurements, interventions, and outcomes.
- **Model how patient recovery evolves over time.**
- **Learn intervention policies** that balance immediate and long-term outcomes in a simplified synthetic environment.

The project is divided into milestones, each introducing a new modeling tool. Throughout these milestones, you will explore different and complementary perspectives on the rehabilitation problem, from data exploration and probabilistic modeling to temporal modeling and sequential decision-making:

- **M0:** Exploratory Data Analysis (**Deadline: 1/October**)
- **M1:** Mixture Models (**Deadline: 11/October**)
- **M2:** Bayesian Networks (**Deadline: 22/October**)
- **M3:** Inference in Bayesian Networks (**Deadline: 2/November**)
- **M4:** Learning Bayesian Networks from Data (**Deadline: 13/November**)
- **M5:** Hidden Markov Models (**Deadline: 24/November**)
- **M6:** Reinforcement Learning (**Deadline: 4/December**)

By the end of this project, you will have explored a range of complementary approaches for modeling rehabilitation under uncertainty: starting with exploratory analysis, uncovering hidden structure with mixture models, capturing probabilistic dependencies with Bayesian networks, reasoning with inference, learning models from data, modeling temporal dynamics with hidden Markov models, and finally learning intervention policies with reinforcement learning.

The dataset used in this project is entirely synthetic and does not represent a particular injury, treatment protocol, or group of real patients. Instead, it provides a simplified representation of recovery following a lower-limb musculoskeletal injury, designed to capture relevant characteristics such as patient heterogeneity, uncertainty, temporal evolution, and sequential interventions. The goal of the project is not only to apply a range of machine learning techniques, but also to understand how they address different and complementary aspects of reasoning, learning, and decision-making under uncertainty.

2. Submission Guidelines

Each group should submit the report for each milestone written in Jupyter Notebook with the name **MX_GYY.ipynb**,

where **X** is the number of the milestone and **YY** is the group number registered in Moodle.

Each submitted notebook should include (but not be limited to):

- The identification of the members of the group (number and name of each element).
- A brief statement on the use of generative AI tools, indicating whether they were used and, if so, for what purposes (e.g., code generation, debugging, clarification of concepts, data analysis, or improvement of written text).
- The code required to perform the analyses, together with the corresponding outputs.
- Clear explanations and justification of relevant modeling and methodological decisions.
- Appropriate presentation and discussion of the results.
- Answers to all tasks, questions, and discussion points specified in the corresponding milestone.

Beware that **I will not run the code of all groups, but I might run some randomly selected groups, so I need explicit outputs in the report** to confirm your conclusions.

Each milestone has its own deadline, and the submission of the notebook should be made by the date at **23:59 in Moodle**. The link to submit each milestone is located in the corresponding week of the deadline.

2.1. Use of Generative AI

The use of generative AI tools (e.g., ChatGPT, Copilot, Gemini) is allowed as a support tool during the development of the project. These tools may be used, for example, to assist with code generation or debugging, clarify concepts, support data analysis, or improve the writing and presentation of the report. However, they should not replace the group's own reasoning, analysis, or understanding of the methods used.

Students are fully responsible for all submitted content, including AI-generated code or text. Any AI-generated output should be critically evaluated and verified. Students should understand and be able to explain the code, methodological choices, results, and conclusions included in their submission.

Errors resulting from the use of generative AI tools, as well as an inability to understand, explain, or adequately justify content produced with their assistance, **may result in a grade penalty**. The penalty will depend on the nature and relevance of the issue identified and may include the assignment of **zero marks to the affected component or components of the work**.

Each submitted notebook should include a brief statement indicating whether generative AI tools were used in the milestone. If generative AI tools were used, the statement should briefly describe the type of assistance provided (e.g., code generation, debugging, clarification of concepts, support with data analysis, improvement of written text); individual prompts or AI-generated code/text do not need to be documented. **If no generative AI tools were used, this should also be explicitly stated. Failure to include this statement will result in a penalty to the corresponding milestone.**

3. Project Goals

The overarching goal of this project is to give you hands-on experience with **probabilistic modeling, inference, temporal modeling, and reinforcement learning** in the context of a longitudinal rehabilitation problem following lower-limb musculoskeletal injury.

By the end of the project, you should not only understand the algorithms covered in class but also appreciate how they can be applied to address different aspects of the same problem.

Concretely, this project will help you:

- **Explore and understand data:** Perform exploratory analysis on longitudinal rehabilitation data, identifying patterns in patient characteristics, recovery states, functional measurements, interventions, and outcomes.
- **Discover hidden structure:** Use mixture models to cluster patient-week observations and investigate latent groupings that may correspond to different recovery profiles.
- **Model dependencies:** Design Bayesian networks that represent probabilistic relationships among patient characteristics, recovery states, observable measurements, interventions, and outcomes.
- **Reason under uncertainty:** Apply exact and approximate inference methods to answer rehabilitation-related probabilistic queries, such as estimating a patient's recovery state from observed measurements.
- **Learn from data:** Use parameter estimation and structure-learning algorithms to learn Bayesian networks from data and compare them with hand-designed models.
- **Capture temporal dynamics:** Fit hidden Markov models to patient trajectories, investigating how underlying recovery states evolve over time and how observable measurements reflect these states.

- **Make decisions:** Apply reinforcement learning to learn intervention policies that account for both immediate utility and long-term recovery within a simplified synthetic environment.

By completing this project, you will gain practical experience with a sequence of machine learning methods that complement each other. More importantly, you will see how these methods provide complementary perspectives on reasoning, learning, and decision-making under uncertainty.

Although the dataset has been simplified for educational purposes, the project illustrates challenges that arise when modeling longitudinal healthcare data, including patient heterogeneity, noisy observations, latent recovery states, temporal evolution, and sequential rehabilitation decisions.

4. Dataset

4.1. Rehabilitation Scenario

Imagine a rehabilitation setting in which patients are followed for several weeks after a lower-limb musculoskeletal injury. At the beginning of rehabilitation, patients differ in characteristics such as age, body mass index, injury severity, comorbidities, and previous physical activity. These factors may influence both their initial functional condition and how their recovery progresses over time.

During rehabilitation, patients are assessed weekly using functional and patient-reported measurements such as mobility, pain, fatigue, functional confidence, activity, and adherence. Their recovery condition evolves over time: some patients improve steadily, others recover more slowly, and some may experience temporary setbacks.

Different rehabilitation interventions may be applied throughout this process, ranging from rest and light exercise to standard or intensive therapy. The effect of an intervention may depend on the patient's current recovery condition, while more intensive rehabilitation may also involve greater short-term burden. Consequently, decisions that appear favorable in the short term may not necessarily lead to the best long-term recovery trajectory.

This creates a **longitudinal dataset** containing both static patient characteristics and dynamic information about the rehabilitation process. Although baseline characteristics remain fixed, recovery state, functional measurements, interventions, and outcomes may change from week to week.

The data provided in this project simulate this rehabilitation setting. Each patient has an underlying recovery condition that evolves throughout the rehabilitation period. Observable measurements provide indirect information about this condition. For the purposes of this synthetic dataset, the underlying recovery state is also provided, allowing different modeling approaches to be developed and evaluated throughout the project.

The dataset does not correspond to a particular lower-limb injury or rehabilitation protocol. Instead, it provides a simplified representation of recovery following lower-limb musculoskeletal injury, designed specifically to support the study of probabilistic modeling, temporal dynamics, and sequential decision-

making.

4.2. Rehabilitation Dataset

The dataset has been synthetically generated to represent longitudinal rehabilitation trajectories following lower-limb musculoskeletal injury. Although the variables and relationships are designed to be plausible, the data do not represent real patients or a particular clinical population.

The dataset consists of two main files: `patients.csv` and `rehabilitation.csv`.

`patients.csv` contains **static characteristics** recorded at baseline, at the beginning of the rehabilitation period. These variables remain fixed over time. The available features are:

- `patient_id`: unique identifier for each patient.
- `age`: age of the patient.
- `sex`: sex of the patient.
- `bmi`: body mass index of the patient.
- `injury_severity`: severity of the lower-limb musculoskeletal injury (Mild, Moderate, or Severe).
- `baseline_mobility`: mobility level at the beginning of rehabilitation, measured on a 0–100 scale, with higher values indicating better mobility.
- `baseline_pain`: pain level at the beginning of rehabilitation. Score from 0 to 10, with higher values indicating higher levels of pain.
- `comorbidity_score`: numerical score summarizing the patient's comorbidity burden. Score from 0 to 5, with higher values indicating a greater comorbidity burden.
- `physical_activity`: physical activity level before the injury (Low, Moderate, or High).
- `risk_score`: synthetic index summarizing baseline factors associated with a less favorable recovery trajectory, expressed on an approximately 0–100 scale, with higher values indicating greater recovery risk.

`rehabilitation.csv` contains **weekly information** collected throughout the rehabilitation period. Each row corresponds to a patient-week pair, allowing the study of temporal recovery dynamics. The available variables are:

- `patient_id`: patient identifier linking the weekly observations to the baseline patient characteristics.
- `week`: rehabilitation week (from 0 to 13).
- `state`: underlying recovery state of the patient during that week. Possible values are Limited, Recovering, and Functional. Although such a state would not be directly observed in practice, it is provided here as the underlying latent state used to generate the observations.
- `mobility`: numerical measure of the patient's mobility. Score from 0 to 100, with higher values indicating better mobility.

- **pain:** numerical measure of the patient's pain. Score from 0 to 10, with higher values indicating higher levels of pain.
- **fatigue:** numerical measure of the patient's fatigue. Score from 0 to 10, with higher values indicating greater fatigue.
- **functional_confidence:** numerical measure representing the patient's confidence in performing functional activities. Score from 0 to 100, with higher values indicating greater functional confidence.
- **activity:** numerical measure of the patient's physical activity during the week. Score from 0 to 110, with higher values indicating greater physical activity.
- **adherence:** numerical measure of the patient's adherence to the rehabilitation program, expressed on a 0–100 scale, with higher values indicating greater adherence.
- **intervention:** rehabilitation intervention applied during that week. Possible values are Rest/Recovery, Light Exercise, Standard Therapy, and Intensive Therapy.
- **utility:** numerical score summarizing the patient's overall rehabilitation outcome during that week. Higher values correspond to more favorable outcomes, taking into account functional status, symptoms, improvement, adherence, and the burden associated with rehabilitation.

This dataset is intended to represent a simplified rehabilitation setting in which patient characteristics, recovery, observable measurements, interventions, and outcomes interact over time. Its probabilistic and temporal structure allows experimentation with clustering, Bayesian networks, hidden Markov models, and reinforcement learning while remaining simple enough for educational purposes.

Why does the dataset include utility? Each patient-week contains a variable called `utility`, which provides a simplified numerical summary of the outcome associated with that week. Utility is not a clinical measurement that would normally be directly recorded in this form. Instead, it is a synthetic quantity that combines different aspects of the rehabilitation process, including functional status, symptoms, improvement, adherence, and the burden associated with the intervention. Later in the project, this variable will serve as the reward signal for reinforcement learning, allowing intervention policies to be evaluated in terms of long-term cumulative utility.

4.3. Limitations of the Dataset

While the dataset is designed to represent plausible rehabilitation trajectories, it is important to recognize its limitations:

- **Synthetic generation:** All patient characteristics, recovery trajectories, measurements, interventions, and outcomes are simulated. Although designed to exhibit plausible relationships, they cannot capture the full complexity of real rehabilitation.
- **Generic injury scenario:** The dataset represents recovery following a lower-limb musculoskeletal injury but does not correspond to a specific injury, diagnosis, or rehabilitation protocol. Different lower-limb injuries can have substantially different recovery processes.

- **Simplified recovery progression:** Recovery is represented by only three discrete states (Limited, Recovering, and Functional). In reality, recovery is continuous, multidimensional, and patient-specific.
- **Simplified observations:** Patient condition is summarized using a small set of weekly measurements. Real rehabilitation data may contain many additional clinical, functional, behavioral, and contextual variables.
- **Restricted interventions:** Only four rehabilitation interventions are considered. Real rehabilitation programs involve more diverse and individualized treatment decisions, and their appropriateness depends on the specific injury and patient.
- **Utility as a proxy:** The utility variable condenses several aspects of rehabilitation into a single numerical value. Real patient outcomes are multidimensional and cannot generally be represented by a single quantity.
- **Simplified temporal dynamics:** Patients are observed at regular weekly intervals over 14 weeks. Real rehabilitation trajectories may have different durations, irregular assessment times, missing observations, and more complex dynamics.
- **No causal or clinical validity:** Relationships in the dataset, including associations between interventions and recovery, should not be interpreted as evidence about the effectiveness of real rehabilitation interventions. The dataset is intended solely as a teaching tool for studying machine learning methods.

Despite these limitations, the dataset is rich enough to support the main goals of the project: exploring how machine learning and probabilistic models can represent uncertainty, uncover latent recovery patterns, model temporal progression, and investigate sequential decision-making in a controlled environment.

Mo: Exploratory Data Analysis (Deadline: 1/October 23:59)

Before we can build probabilistic models, it is important to familiarize ourselves with the dataset and the type of information it contains. In this milestone, your task is to conduct a brief exploratory data analysis (EDA) to gain a deeper understanding of the data structure and formulate hypotheses about the relationships between patient characteristics, recovery states, functional measurements, rehabilitation interventions, and outcomes.

You will work with the two provided files:

- `patients.csv` (static baseline characteristics for each patient);
- `rehabilitation.csv` (longitudinal weekly rehabilitation data for each patient, including recovery state, functional measurements, interventions, and utility).

For this milestone, **treat each patient-week (row in `rehabilitation.csv`) as an independent observation** when analyzing recovery states, measurements, interventions, and utility, **except when explicitly analyzing changes over time or individual patient trajectories**. Use `patients.csv` for static baseline

summaries.

Your analysis should include both summary statistics and visualizations. You do not need to be exhaustive; the goal is to uncover patterns that will motivate the probabilistic models in later milestones. The focus here is **interpretation**: try to reason about what the patterns you observe might mean in a rehabilitation context, while remembering that the data is synthetic.

Concretely, you should address the following 6 tasks:

Task 1: Dataset overview

- How many patients are included? How many patient-week observations are there in total?
- How many weeks are available per patient on average?
- Are all patients observed for the same number of weeks?
- Briefly identify which variables are static patient characteristics and which vary over time.

Task 2: Baseline patient characteristics

Using only `patients.csv`:

- What are the distributions of age, sex, BMI, injury severity, comorbidity score, and physical activity?
- What are the distributions of baseline mobility and baseline pain?
- How does the risk score distribute across the population?
- Are there noticeable relationships between risk score and baseline characteristics such as injury severity, comorbidity score, or baseline mobility?

Task 3: Recovery states and trajectories

Using `rehabilitation.csv`:

- What proportion of patient-weeks belong to each recovery state (Limited, Recovering, Functional)?
- How does the proportion of patients in each state change over time?
- Do all patients follow similar recovery trajectories, or do some recover more quickly, remain stable, or experience setbacks? Provide a few illustrative examples.

Task 4: Functional and patient-reported measurements

- Summarize the distributions of mobility, pain, fatigue, functional confidence, activity, and adherence.
- Do these variables appear to differ across recovery states?
- How do selected measurements evolve over the rehabilitation period? Choose appropriate variables and visualizations to illustrate the main patterns you observe.

Task 5: Rehabilitation interventions and outcomes

- What is the distribution of rehabilitation interventions across patient-weeks?
- How do interventions vary across recovery states?
- What is the distribution of utility values?
- Does utility differ systematically across recovery states or interventions?

- Based only on this exploratory analysis, can you conclude which intervention is best for a patient in a particular state? Explain why or why not.

Task 6: Data quality check

- Are there any missing values or duplicated patient-week observations?
- Are the values and ranges of the variables consistent with their descriptions?
- Identify any other potential data-quality issues that should be considered before applying machine learning methods.

The emphasis of this milestone is **not on sophisticated analysis**, but rather on developing an intuitive understanding of the data that will guide the modeling decisions made in subsequent milestones.

M1: Mixture Models (Deadline: 11/October 23:59)

In this milestone, you will use **mixture models** to uncover hidden structure in the population of patient-week rehabilitation observations. Each row in `rehabilitation.csv` should be treated as an independent observation, since functional measurements and rehabilitation outcomes may change over time even for the same patient.

Here, you will **use observed rehabilitation measurements to cluster patient-weeks**, with the goal of exploring the overall structure of the data and investigating whether latent groups correspond to different recovery states. Later milestones will focus on other aspects of the problem, for example, in M5 we will explicitly consider the temporal ordering of observations using hidden Markov models.

By fitting a **Gaussian mixture model (GMM)**, we can explore whether latent groups capture patterns similar to the recovery states or reveal alternative recovery profiles. The goal of this milestone is not to achieve perfect alignment between clusters and recovery states, but rather to practice fitting mixture models and interpreting latent group structure in rehabilitation data. The clusters may or may not match the labeled states; this is part of the analysis.

Concretely, you should address the following 4 tasks:

Task 1: Feature selection and preprocessing

- Choose the features from `rehabilitation.csv` that you want to use for clustering. Remember that each row (patient-week) should be treated as an independent observation.
- Briefly justify your choice of features.
- Standardize or normalize the selected features so that features measured on different scales are comparable.

Remark: Do not use `state`, `intervention`, or `utility` when fitting the mixture model. These should not be used as indicators for clustering: `state` is the hidden ground truth, `intervention` is a decision variable, and `utility` is an outcome (reward) variable. The goal of clustering is to discover latent structure using only observable indicators of the patient's

rehabilitation status. These excluded features may later be used to interpret the clusters, but they should not be included when fitting the model.

Task 2: Model fitting and selection

- Fit **Gaussian mixture models (GMMs)** with different numbers of components (for example, from 2 to 8).
- Use information criteria (AIC and/or BIC) to compare the models and select a reasonable number of components.
- For the selected model, assign each patient-week to its most likely component.
- Examine the posterior membership probabilities. Are most patient-weeks assigned to one component with high probability, or are some observations ambiguous?

Task 3: Cluster characterization

- Characterize the resulting clusters using the rehabilitation measurements included in the model. Use appropriate summary statistics and visualizations.
- Identify the main characteristics that distinguish the clusters. For example, do some clusters correspond to patient-weeks with better mobility and lower pain, while others show greater functional limitation or symptom burden?
- Compare the clusters with the true recovery states (*Limited*, *Recovering*, and *Functional*). Do the clusters align with these states, or is there substantial overlap?
- Identify which recovery states are most frequently mixed within the same cluster and discuss possible reasons for this overlap.

Task 4: Interpretation

- Give a rehabilitation interpretation to each cluster based on its observed characteristics.
- Examine whether *intervention* and *utility* differ across the identified clusters. Remember that these variables were not used to fit the GMM.
- Provide at least one table or figure illustrating the relationship between clusters and recovery states.
- Discuss whether the clusters appear primarily to represent recovery levels or whether they reveal additional patterns not completely captured by the three recovery states.
- Briefly discuss the limitations of treating patient-weeks as independent observations. What information is being ignored by the GMM?

M2: Bayesian Networks (Deadline: 22/October 23:59)

In the previous milestone, you used mixture models to explore hidden groupings of patient-week rehabilitation observations based on functional and patient-reported measurements. This revealed possible structures in the data, but it did not explicitly model how different variables depend on each other.

In this milestone, you will design a **Bayesian network (BN)** to represent probabilistic dependencies among patient characteristics, recovery states, observable rehabilitation measurements, interventions,

and outcomes. Unlike clustering, a BN explicitly represents assumptions about the *direction of dependencies* (for example, recovery state $\rightarrow$ mobility) and allows you to reason about conditional independencies.

At a minimum, your Bayesian network should include the variables: `risk_score`, `state`, `intervention`, and `utility`, together with several observed rehabilitation measurements. This allows the BN to represent how baseline patient characteristics relate to recovery, how recovery states relate to observable measurements, how rehabilitation interventions are selected, and how these variables relate to outcomes.

The goal is not to build a perfect rehabilitation model, but to practice **formalizing modeling assumptions** and understanding how probabilistic dependencies emerge from a structured graphical representation. In later milestones, you will use the network for inference and compare your hand-designed model with networks learned directly from data.

Concretely, you should address the following 5 tasks:

Task 1: Structure design

- Design a Bayesian network that models relationships among baseline patient characteristics, recovery state, observable rehabilitation measurements, intervention, and utility. At a minimum, your network should contain `risk_score`, `state`, `intervention`, and `utility`, together with several observed variables selected from `mobility`, `functional_confidence`, `pain`, `fatigue`, `activity`, and `adherence`. You may include additional variables from `patients.csv` if they are relevant to your model.
- Draw the corresponding graph and justify your modeling assumptions. For example, recovery state may influence mobility and pain, while patient risk may influence recovery state.
- Implement the structure of the network in `pgmpy`.

Task 2: Discretization

- Continuous variables cannot be used directly in the discrete Bayesian networks considered in this project. Select appropriate thresholds to discretize the continuous variables included in your network into a small number of meaningful categories.
- Document your discretization rules and explain why they are reasonable.

Task 3: Estimating CPTs by counting

- Use the full dataset to approximate the conditional probability tables (CPTs) using relative frequencies (counting occurrences). At this stage, we are more interested in structure than predictive accuracy. Treat each patient-week in `rehabilitation.csv` as an independent observation, so that probabilities reflect week-level rehabilitation patterns rather than patient-level summaries. When using baseline variables from `patients.csv`, merge them with the corresponding patient-week observations using `patient_id`.
- For variables with multiple parents, compute counts conditioned on all combinations of parent values.
- Draw the BN graph (e.g., `pgmpy` plotting utilities or `networkx`).

- Inspect several probability tables and discuss whether they appear consistent with your expectations. For example, depending on your network:
 - Does a high risk score increase the probability of being in the Limited state?
 - Are Functional patients more likely to have high mobility or low pain?
 - Are different interventions more common in different recovery states?
 - How does utility vary with recovery state and intervention?

Task 4: Conditional independence analysis

Using your network:

- Identify several conditional independence relationships implied by your BN. Use pgmpy's `get_independencies()` functionality where appropriate.
- Determine whether selected pairs of variables are d-separated given specific evidence.
- Explain these independencies in the rehabilitation context. For example, depending on your network:
 - Is mobility independent of risk score once recovery state is known?
 - Are mobility and pain independent once recovery state is known?
 - Does conditioning on intervention or utility change dependencies between other variables?

Task 5: Interpretation

Interpret your network from a rehabilitation perspective. Discuss questions such as:

- Which observable measurements appear to provide the most information about recovery state?
- Do the estimated probabilities agree with your expectations and with the exploratory analysis from Mo?
- What assumptions or simplifications are implicit in your network?
- Which relationships in the BN represent modeling assumptions, and which are directly supported by the estimated probabilities?

Reflect on the limitations of your network and on how it differs from a true causal model of rehabilitation.

Remark: There is no single “correct” Bayesian network. Different groups may propose different structures and obtain different probability tables. The purpose of this milestone is not to discover the true rehabilitation process, but rather to understand how graphical models can represent assumptions about dependencies among variables and support reasoning under uncertainty.

M3: Inference in Bayesian Networks (Deadline: 2/November 23:59)

In the previous milestone, you designed a Bayesian network (BN) to capture dependencies among patient characteristics, recovery states, rehabilitation measurements, interventions, and outcomes. Defining the structure and estimating conditional probability tables gave you a static probabilistic model describing how these variables are related.

In this milestone, you will take the next step: **reasoning with the BN by performing inference**. The goal is to use your model to answer meaningful questions about rehabilitation, such as estimating a patient's recovery state given observed measurements, determining which intervention is likely to be applied, or estimating the probability of a favorable outcome given patient characteristics and rehabilitation decisions.

You will begin with **exact inference** using Variable Elimination and Belief Propagation, and then move to **approximate inference** using sampling. This will allow you to compare different approaches to probabilistic inference and explore how different inference methods can be used to answer probabilistic questions under uncertainty.

You will work primarily with the **Bayesian network built in M2** and with the **discretized dataset from M2**.

Concretely, you should address the following 3 tasks:

Task 1: Exact inference

Using `pgmpy`, formulate and compute posterior probabilities for three queries, each addressing one of the following aspects:

- (a) **Recovery-state assessment:** infer the patient's recovery state given one or more observable rehabilitation measurements.
- (b) **Rehabilitation intervention:** infer the distribution of interventions given relevant information about the patient's recovery state and/or observed measurements.
- (c) **Outcome:** infer the distribution of utility given relevant patient characteristics, recovery information, and/or intervention.

For each query:

- Clearly state the probabilistic query and explain the rehabilitation question it represents.
- Use both Variable Elimination and Belief Propagation to compute the posterior distribution.
- Compare the posterior distributions obtained with both methods. The results should be nearly identical. If differences appear, discuss possible reasons.
- Interpret the resulting posterior probabilities in the context of rehabilitation.

Task 2: Approximate inference

Implement **one sampling-based inference method**. For the same queries considered in Task 1:

- compute posterior distributions using approximate inference;
- compare them to the exact posterior distributions;
- study how the approximation changes as the number of samples increases;
- discuss the trade-off between computational efficiency and accuracy.

For queries involving a complete distribution, select at least one specific value of the query variable when quantitatively comparing exact and approximate probabilities.

Task 3: Evidence and probabilistic reasoning

Select one or two of the questions investigated in Task 1 and study how the posterior distribution changes as different evidence is introduced. For example, you may:

- Start from the prior distribution of state and progressively introduce evidence about mobility, pain, or other measurements.
- Compare the inferred intervention distribution for patients with different recovery states or observed measurements.
- Compare the probability of a favorable outcome under different combinations of evidence.

Discuss which evidence has the greatest effect on the posterior distribution and explain the results using the dependencies encoded in your Bayesian network.

M4: Learning Bayesian Networks from Data (Deadline: 13/November 23:59)

In the previous milestones, you designed a Bayesian network (BN) based on assumptions about the rehabilitation process and used it to perform probabilistic inference. This demonstrated how a hand-designed model can represent dependencies and support reasoning under uncertainty.

In this milestone, you will explore an alternative approach: **learning Bayesian networks directly from data**. Instead of relying exclusively on modeling assumptions, you will allow algorithms to estimate parameters of your network and to discover network structures automatically.

This will allow you to compare:

- how well the **hand-designed BN** from M2 represents your assumptions about rehabilitation, and
- how well a **data-driven BN** learned automatically represents the observed patient-week data.

You will begin by learning parameters for your hand-designed BN and then move to **structure learning** using score-based search. Finally, you will evaluate and compare the resulting models on unseen patients, reflecting on the trade-off between **interpretability** (expert-designed networks) and **fit to data** (learned networks).

For this milestone, continue to use the discretized variables and discretization rules defined in M2. Both parameter learning and structure learning should be performed on the corresponding discretized dataset.

Concretely, you should address the following 4 tasks:

Task 1: Parameter learning

- Split the discretized data into training and test sets. The split should be performed at the **patient level**, not at the patient-week level:
 - assign 70% of patients to the training set and 30% to the test set;
 - use all weeks from each patient consistently in either the training or test set.

This prevents information leakage between training and test sets and evaluates the models on patients that were not used during training.

- Start with the BN structure designed in M2.
- Use a parameter estimator of your choice, such as Maximum Likelihood Estimation, or Bayesian Estimation, to learn the CPTs from the training data.
- Compare selected learned CPTs with the frequency-based CPTs obtained in M2. Discuss:
 - Are the estimated probabilities similar?
 - What happens to probabilities associated with configurations that were rare in the data?
 - If you use Bayesian estimation, how do the resulting CPTs differ from those obtained by simple counting?

Task 2: Structure learning

- Use a score-based structure-learning algorithm to learn a BN structure directly from the training data. Use the same set of variables as in your hand-designed BN to facilitate comparison. **Do not impose constraints on the network structure.**
- Visualize the learned network.
- Compare the learned structure with the network designed in M2. Discuss:
 - Which dependencies are present in both networks?
 - Which dependencies are present only in one of the networks?
 - Are any relationships represented with the opposite direction?
 - Did the algorithm identify dependencies that you did not include in your hand-designed model?
- Critically assess the learned structure in the rehabilitation context. Are all the learned relationships and their directions meaningful? Identify any relationships that appear difficult to justify from a domain perspective.

Task 3: Model comparison

- Evaluate the hand-designed BN and the automatically learned BN on the test data using an appropriate scoring criterion, such as log-likelihood or BIC. You may consider more than one score, but select one criterion for your main comparison.
- Discuss:
 - Which model provides a better statistical fit to the held-out data?
 - How different are the two networks structurally?
 - Are relationships in the data-driven network as easy to interpret as those in the hand-designed network?
 - What do these results tell you about the difference between designing a BN using domain assumptions and learning one from observational data?

Task 4: Reflection

In practice, model evaluation could be made more robust using cross-validation.

- What advantages could cross-validation provide when evaluating Bayesian networks?

- What additional considerations arise because each patient contributes multiple weekly observations?

Purely data-driven structure learning identifies statistical dependencies, which do not necessarily represent causal relationships. Write a short discussion addressing the following question:

How would you combine knowledge about rehabilitation with data-driven methods when designing a Bayesian network?

For example, consider whether some relationships should be required or prohibited based on domain knowledge, whether some edge directions are more meaningful than others, and how expert knowledge and automatic structure learning could complement each other.

M5: Hidden Markov Models (Deadline: 24/November 23:59)

So far, we have treated each patient-week as an independent observation. In reality, however, patients are followed throughout rehabilitation, and their underlying recovery condition evolves over time. To capture this temporal dependence, we will use **hidden Markov models (HMMs)**.

In an HMM, we assume that:

- each patient has a **hidden recovery state** that evolves over time according to Markov dynamics;
- at each week, we observe rehabilitation measurements that depend probabilistically on the current hidden state.

In principle, an HMM could incorporate several observable variables. However, for this milestone, you will use **two rehabilitation measurements**. This keeps the model simple and interpretable while still allowing meaningful recovery dynamics to be captured.

You will work with the **train and test split from M4**.

This milestone introduces temporal models of recovery progression, showing how to perform inference in sequential data and learn model parameters from patient trajectories.

Task 1: Model setup

- Treat the **true variable** state in `rehabilitation.csv` as hidden (ignore it during training).
- Select **two observable rehabilitation variables** that you consider informative about the patient's recovery state. Briefly justify your choice.
- Based on the type of observations you selected, choose an appropriate HMM emission model among those studied in the laboratory and justify your choice.
- Perform any preprocessing required by the selected model. When preprocessing requires estimating parameters from the data, use only the training set.
- Organize the data so that each patient corresponds to one sequence of observations, ordered by week.

Task 2: Parameter learning

- Use the **Baum-Welch algorithm**, implemented in `hmmlearn`, to estimate the HMM parameters.
- Fit an HMM with **3 hidden states**, matching the number of underlying recovery states in the dataset. Run the algorithm several times with different random initializations and retain an appropriate fitted model based on the training data.
- Inspect the learned:
 - initial-state probabilities;
 - transition probabilities;
 - emission distributions.

The learned hidden states are unlabeled: the algorithm may identify them simply as State 0, State 1, and State 2. Use the learned emission distributions to interpret the states and propose a correspondence with `Limited`, `Recovering`, and `Functional`.

- After interpreting the states:
 - Examine the transition matrix. Does it represent plausible rehabilitation dynamics? Are patients likely to remain in the same recovery state from one week to the next? Are improvements and setbacks represented?
 - Examine the emission distributions. Are the observable measurements associated with the states in a way that is consistent with your interpretation?
- Finally, discuss why selecting a model solely based on its training likelihood may be problematic and what alternatives could be used.

Task 3: Inference and evaluation

- Apply the **Viterbi algorithm** to obtain the most likely sequence of hidden states for each patient in the test set.
- Determine a mapping between the unlabeled HMM states and the true recovery states (`Limited`, `Recovering`, `Functional`). Clearly explain how you obtained this mapping.
- After defining the mapping, compare the decoded states with the true state variable, which was not used during training.
- Evaluate the results using:
 - a confusion matrix to see where the model makes errors;
 - accuracy of the decoded states;
 - plots comparing the decoded and true state sequences for two or three patients.
- Discuss the errors:
 - Which recovery states are most frequently confused?
 - At which parts of patient trajectories do errors tend to occur?
 - Does the HMM capture gradual recovery and setbacks?
 - What limitations of the HMM assumptions might explain the observed errors?

M6: Reinforcement Learning (Deadline: 4/December 23:59)

Up to this point, you have built models to analyze and describe different aspects of patient recovery:

- Mixture models revealed hidden structure in patient-week rehabilitation observations.
- Bayesian networks captured probabilistic dependencies among patient characteristics, recovery states, measurements, interventions, and outcomes.
- Hidden Markov models described how recovery states evolve over time.

In practice, however, rehabilitation also involves sequential decisions about how to support a patient's recovery over time. This brings us to the final milestone: **Reinforcement Learning (RL)**.

In this milestone, you will view rehabilitation as a sequential decision-making problem:

- The **environment** represents the patient's rehabilitation trajectory.
- The **state** is the patient's recovery state (hidden in reality, but available in our synthetic dataset).
- The **actions** are the possible rehabilitation interventions (Rest/Recovery, Light Exercise, Standard Therapy, Intensive Therapy).
- The **reward** is derived from the **utility**, which provides a simplified summary of the rehabilitation outcome associated with each patient-week.

You will implement **tabular Q-learning** to learn an intervention policy that maximizes long-term utility. You will compare the learned policy with simple baseline policies and critically interpret the resulting recommendations.

The agent cannot interact with real patients. Instead, the historical trajectories in the synthetic dataset will be used to estimate a simplified rehabilitation environment. The learned policy should therefore be interpreted as the behavior of an RL agent within this **synthetic, data-derived environment**, rather than as a clinically validated rehabilitation policy. This setup also illustrates a challenge that arises in clinical applications, where experimenting freely with interventions on patients may be impractical or ethically inappropriate.

This final milestone introduces a different perspective on the rehabilitation problem, moving from modeling recovery to sequential decision-making: building not just models that describe patient recovery, but agents that learn how to select actions over time. In real-world settings, RL is applied with much larger state/action spaces, often requiring function approximation (e.g., deep RL). Our toy environment will illustrate the principles in a controlled way.

Task 1: Environment setup

Treat each patient trajectory as a sequence of transitions

$$(s_t, a_t, r_t, s_{t+1})$$

where s_t is the recovery state at week t , a_t is the intervention applied at week t , r_t is the reward, and s_{t+1} is the recovery state at the following week.

Transition probabilities

- Estimate the transition probabilities

$$P(s_{t+1}|s_t, a_t)$$

from the observed patient trajectories using relative frequencies.

- Construct a transition matrix of shape $n_states \times n_actions \times n_states$.
- Visualize the resulting transition probabilities and interpret them in the rehabilitation context. How does the probability of recovery, stability, or setback vary across states and interventions?

Reward function

- Estimate the expected reward $r(s, a)$ as the average utility observed when intervention a is applied in state s

$$r(s, a) = \mathbb{E}[utility|s, a]$$

- Construct and visualize the resulting state-action reward matrix of shape $n_states \times n_actions$.
- Discuss how immediate rewards differ across interventions and recovery states.

Initial state distribution

- Estimate the initial-state distribution π (vector of size n_states) from each patient's first rehabilitation week.

Environment

- Use these estimated transition probabilities, rewards, and initial-state probabilities to construct a simulator in which the next state is sampled from.

```

1  # environment simulator for Q-learning
2  class RehabilitationEnv:
3      def __init__(self, pi, P, R, horizon=13, seed=0):
4          self.pi = pi
5          self.P = P
6          self.R = R
7          self.horizon = horizon
8          self.n_states = P.shape[0]
9          self.rng = np.random.default_rng(seed)
10
11     def reset(self):
12         self.t = 0
13         self.s = int(self.rng.choice(self.n_states, p=self.pi))
14         return self.s
15
16     def step(self, a):
17         reward = float(self.R[self.s, a])
18         s_next = int(self.rng.choice(self.n_states,
19                                     p=self.P[self.s, a]))
20         self.s = s_next
21         self.t += 1
22         done = self.t >= self.horizon
23         return s_next, reward, done, {}
24
25     env = RehabilitationEnv(pi, trans_mat, r_sa)

```

Task 2: Tabular Q-learning

Implement tabular Q-learning. Choose and justify reasonable values for:

- learning rate α ;
- discount factor γ ;
- exploration parameter ϵ .

Train the agent by interacting with the rehabilitation environment constructed in Task 1. After training:

- Display the learned Q-table.
- Derive the corresponding policy.
- Identify the preferred intervention for each recovery state.
- Interpret the resulting policy in the rehabilitation context.
- Investigate how the learned policy changes for different values of the discount factor γ .
What does this reveal about the trade-off between immediate and long-term utility?

Task 3: Baseline comparison

Construct two baseline policies:

- **Random policy:** select each intervention uniformly at random.
- **Heuristic policy:** design a simple rehabilitation policy based on your understanding of the problem. Explain the rationale behind your choices.

Evaluate the random, heuristic, and learned Q-learning policies in the simulated environment over multiple episodes. Use the function below to evaluate any policy. This function returns the cumulative returns obtained over 1000 episodes.

Compare:

- cumulative reward;
- average reward;
- variability of returns;
- interventions selected by each policy.

Discuss the differences among the policies.

```

1  # --- Generic simulator for any policy ---
2  def evaluate_policy(env, policy_fn, Q=None, episodes=1000, seed=0):
3      rng = np.random.default_rng(seed)
4      returns = []
5      for _ in range(episodes):
6          s = env.reset()
7          G = 0.0
8          for t in range(env.horizon):
9              a = policy_fn(s, Q=Q, rng=rng)
10             s, r, done, _ = env.step(a)
11             G += r
12             if done: break
13             returns.append(G)
14     return np.array(returns)

```

Task 4: Policy analysis

Analyze the behavior of the learned policy.

- Which interventions are preferred for each recovery state?
- Are some interventions rarely selected?
- How does the learned policy differ from the immediate reward matrix examined in Task 1?
- Are there situations in which an intervention with lower immediate reward is preferred because of its effect on future recovery?
- How sensitive is the policy to the discount factor?
- Are any learned decisions surprising from a rehabilitation perspective?

Discuss possible explanations for unexpected results, considering:

- the estimated transition probabilities;
- the reward definition;
- stochasticity and noise in the data;
- exploration during learning;
- limitations of the synthetic environment.

In particular, discuss the difference between **maximizing immediate utility** and **maximizing long-term cumulative utility**.

Task 5: Reflection

Discuss the potential and limitations of applying reinforcement learning to rehabilitation decision-making. Consider questions such as:

- What advantages can sequential decision-making offer compared with selecting interventions independently at each week?
- What challenges arise when rewards are delayed, noisy, or only imperfect proxies for patient outcomes?
- Why should a policy learned from observational trajectories not automatically be interpreted as evidence that one intervention is clinically preferable to another?
- How could clinical expertise and RL-based decision-support systems complement each other?

Reflect on the simplifications of the current setting:

- only three recovery states;
- only four interventions;
- utility as a simplified reward signal;
- a small Markov state representation;
- transition and reward functions estimated from synthetic historical trajectories;
- no interaction with real patients.

How would a real rehabilitation decision-support problem differ from the framework considered in this project?